GUIDE 06

# Cost Analysis: Pay-As-You-Go vs PTU

Enterprise-scale pricing comparison for Azure OpenAI and LLM API consumption

**Agentic AI Builder Expert Bootcamp**

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# 1. Understanding the Two Pricing Models

Azure OpenAI Service offers two fundamentally different pricing models. Choosing the right one can save enterprises tens of thousands of dollars per month.

| Dimension | Pay-As-You-Go (PAYG) | Provisioned Throughput (PTU) |
|---|---|---|
| Billing basis | Per token consumed | Per unit of reserved capacity per hour |
| Commitment | None (pay per use) | Monthly or annual reservation |
| Latency | Variable (shared infra) | Consistent (dedicated capacity) |
| Throughput | Subject to rate limits | Guaranteed minimum throughput |
| Cost predictability | Variable (depends on usage) | Fixed monthly cost |
| Scale-to-zero | Yes (no usage = no cost) | No (you pay even if idle) |
| Best for | Variable/unpredictable workloads | Steady, high-volume production |

**Simple analogy:** PAYG is like a taxi (pay per ride, no commitment). PTU is like leasing a company car (fixed monthly payment, always available, worth it if you drive enough).

# 2. Pay-As-You-Go (PAYG) Deep Dive

## How It Works

- You are charged per 1,000 tokens (roughly 750 words) processed.
- Input tokens (your prompt) and output tokens (model response) are priced differently.
- No upfront commitment. Scale up or down instantly.
- Shared infrastructure means latency can vary during peak times.

## Token Pricing (Azure OpenAI, as of Early 2026)

| Model | Input (per 1M tokens) | Output (per 1M tokens) | Context Window |
|---|---|---|---|
| GPT-4o | $2.50 | $10.00 | 128K |
| GPT-4o-mini | $0.15 | $0.60 | 128K |
| GPT-4.1 | $2.00 | $8.00 | 1M |
| GPT-4.1-mini | $0.40 | $1.60 | 1M |
| GPT-4.1-nano | $0.10 | $0.40 | 1M |
| o3-mini | $1.10 | $4.40 | 200K |

> ***Warning:*** *Token pricing changes frequently. Always verify current pricing on the Azure OpenAI pricing page before making decisions.*

### Rate Limits

PAYG comes with default rate limits that can bottleneck high-traffic applications:

- Tokens per minute (TPM): 60K-800K depending on model and region
- Requests per minute (RPM): 60-3,500 depending on model
- Can request increases via Azure support, but not guaranteed

## 3. Provisioned Throughput Units (PTU) Deep Dive

### How It Works

- You reserve a fixed number of PTUs (Provisioned Throughput Units).
- Each PTU provides a guaranteed amount of processing capacity measured in tokens per minute.
- You get dedicated compute infrastructure -- consistent, low latency.
- Billed hourly regardless of actual usage.

### PTU Capacity Estimates

The number of tokens a PTU can process depends on the model and prompt/completion ratio. Here are approximate values:

| Model | Approx. TPM per PTU | Min PTUs | Monthly Cost per PTU (approx.) |
|---|---|---|---|
| GPT-4o | ~2,500 TPM | 50 | $1,200 - $2,000 |
| GPT-4o-mini | ~10,000 TPM | 25 | $300 - $600 |
| GPT-4.1 | ~2,000 TPM | 50 | $1,500 - $2,500 |

> ***Warning:*** *PTU pricing and capacity vary by region and are subject to change. These are approximate figures for planning. Use the Azure PTU calculator for exact numbers.*

### Reservation Options

- **Monthly reservation:** Higher per-unit price, more flexibility. Cancel with 30-day notice.
- **Annual reservation:** ~30-40% discount over monthly. Requires 1-year commitment.
- **Hourly (limited availability):** Pay-as-you-go PTU pricing for burst capacity.

## 4. When to Use Which Model

| Scenario | Recommended | Reasoning |
|---|---|---|
| Startup / MVP / POC | PAYG | Unpredictable usage, need to minimize fixed costs |
| Dev/test environments | PAYG | Low, intermittent usage. Scale to zero. |
| Internal chatbot (50 users) | PAYG | Moderate usage, likely under PTU break-even |
| Customer-facing agent (1000+ DAU) | PTU | High, predictable volume; latency matters |
| Batch processing (nightly) | PAYG (Batch API) | 50% discount on batch; no latency needs |
| Real-time trading agent | PTU | Latency consistency is critical |
| Multi-model routing | PAYG | Flexibility to switch models without commitment |
| Regulated industry (SLA needed) | PTU | Dedicated capacity = guaranteed availability |

# 5. Cost Calculation Framework

## PAYG Monthly Cost Formula

```
PAYG Monthly Cost =
  (Monthly Input Tokens / 1,000,000) x Input Price per 1M
+ (Monthly Output Tokens / 1,000,000) x Output Price per 1M

Example (GPT-4o):
  Input:  50M tokens/month x $2.50/1M  = $125.00
  Output: 20M tokens/month x $10.00/1M = $200.00
  Total:  $325.00/month
```

## PTU Monthly Cost Formula

```
PTU Monthly Cost =
  Number of PTUs x Cost per PTU per hour x 730 hours/month

Example (GPT-4o, 100 PTUs):
  100 PTUs x ~$2.00/hour = $200/hour
  $200/hour x 730 hours = $146,000/month

Capacity provided:
  100 PTUs x 2,500 TPM = 250,000 TPM
  250,000 TPM x 60 min x 730 hours = ~10.95 billion tokens/month
```

***Tip:*** *The key insight: PTU becomes cheaper than PAYG at high, sustained volumes. The crossover point depends on your model, usage pattern, and ratio of input to output tokens.*

# 6. Scenario Analysis: Small Scale

**Scenario:** Internal support chatbot. 50 employees, ~20 queries/day each, average 500 input + 300 output tokens per query.

```
Daily queries: 50 users x 20 queries = 1,000 queries/day
Monthly queries: 1,000 x 22 workdays = 22,000 queries/month

Input tokens:  22,000 x 500  = 11M tokens/month
Output tokens: 22,000 x 300  = 6.6M tokens/month

PAYG Cost (GPT-4o):
  Input:  11M / 1M x $2.50  = $27.50
  Output: 6.6M / 1M x $10   = $66.00
  Total: $93.50/month

PAYG Cost (GPT-4o-mini):
  Input:  11M / 1M x $0.15  = $1.65
  Output: 6.6M / 1M x $0.60 = $3.96
  Total: $5.61/month

PTU Cost (GPT-4o, min 50 PTUs):
  ~$60,000+/month

Verdict: PAYG wins overwhelmingly.
```

# 7. Scenario Analysis: Medium Scale

**Scenario:** Customer-facing AI assistant. 5,000 daily active users, 10 queries each, 800 input + 500 output tokens average. Using GPT-4o.

```
Daily queries: 5,000 x 10 = 50,000 queries/day
Monthly queries: 50,000 x 30 = 1,500,000 queries/month

Input tokens:  1.5M x 800  = 1.2B tokens/month
Output tokens: 1.5M x 500  = 750M tokens/month

PAYG Cost (GPT-4o):
  Input:  1,200M / 1M x $2.50  = $3,000
  Output: 750M / 1M x $10.00   = $7,500
  Total: $10,500/month

PTU Analysis (GPT-4o):
  Required TPM: Peak load estimate
  50K queries/day / 1440 min = ~35 queries/min
  35 x 1,300 tokens = ~45,500 TPM needed
  PTUs needed: 45,500 / 2,500 = ~18 PTUs
  But minimum is 50 PTUs.

  50 PTUs x $2.00/hr x 730 = $73,000/month

Verdict: PAYG wins. PTU would cost 7x more.
```

# 8. Scenario Analysis: Enterprise Scale

**Scenario:** Enterprise agentic platform. 50,000 daily users, 30 queries each, multi-step agents averaging 5 LLM calls per query, 1,200 input + 800 output tokens per call. Latency SLA required.

```
Daily LLM calls: 50,000 x 30 x 5 = 7,500,000 calls/day
Monthly LLM calls: 7.5M x 30 = 225,000,000 calls/month

Input tokens:  225M x 1,200 = 270B tokens/month
Output tokens: 225M x 800   = 180B tokens/month

PAYG Cost (GPT-4o):
  Input:  270,000M / 1M x $2.50  = $675,000
  Output: 180,000M / 1M x $10.00 = $1,800,000
  Total: $2,475,000/month

PAYG Cost (GPT-4o-mini):
  Input:  270,000M / 1M x $0.15  = $40,500
  Output: 180,000M / 1M x $0.60  = $108,000
  Total: $148,500/month

PTU Analysis (GPT-4o):
  Peak TPM needed (assume 8-hour peak):
  7.5M calls / 480 min = ~15,625 calls/min
  15,625 x 2,000 tokens = ~31.25M TPM
  PTUs: 31,250,000 / 2,500 = ~12,500 PTUs
  Cost: 12,500 x $2.00/hr x 730 = $18,250,000/month

  Annual reservation (~35% discount):
  $18,250,000 x 0.65 = ~$11,862,500/month

Verdict: At this extreme scale, even PAYG for GPT-4o
is $2.5M/month. The real strategy is:
1. Use GPT-4o-mini for most calls ($148K)
2. Reserve PTUs only for GPT-4o on critical paths
3. Implement aggressive caching (30-50% reduction)
4. Use model cascading (cheap model first)
```

# 9. Hidden Costs and Considerations

| Hidden Cost | PAYG Impact | PTU Impact |
|---|---|---|
| Retry tokens (on failures) | You pay for retries too | Absorbed by reserved capacity |
| Prompt engineering overhead | Longer prompts = higher cost | Same, but capacity is fixed |
| Cache misses | Every miss = full token cost | Hits capacity ceiling |
| Idle capacity | None (no usage = no cost) | You still pay for unused PTUs |
| Burst capacity | Rate-limited, may queue | Guaranteed within capacity |
| Regional pricing | Varies by Azure region | Varies by region + availability |

| Hidden Cost | PAYG Impact | PTU Impact |
|---|---|---|
| Data transfer | Standard Azure egress fees | Same |
| Monitoring tools | LangSmith/Helicone costs | Same |

# 10. UiPath AI Unit Pricing Comparison

For UiPath customers, AI features are consumed via AI Units. Understanding how these compare to direct API pricing is critical for enterprise budgeting.

## UiPath AI Unit Model

- **AI Units** are a universal credit system for UiPath AI features (Document Understanding, Communications Mining, Autopilot, GenAI activities).
- Each AI action consumes a certain number of AI Units based on the model and action type.
- Purchased in packs (e.g., 1,000 AI Units/month) as part of UiPath licensing.

## Comparison Framework

| Dimension | UiPath AI Units | Direct API (PAYG) | Direct API (PTU) |
|---|---|---|---|
| Pricing unit | AI Unit (abstracted) | Per token | Per PTU/hour |
| Transparency | Bundled, opaque | Fully transparent | Transparent per PTU |
| Model flexibility | UiPath-selected models | Any model you choose | Specific model per PTU |
| Infrastructure | Managed by UiPath | Self-managed | Self-managed |
| Integration effort | Native UiPath connectors | Custom API integration | Custom API integration |
| Vendor lock-in | High (UiPath ecosystem) | Low | Medium (Azure) |
| Best for | UiPath-centric RPA + AI | Custom agent platforms | High-volume, latency-sensitive |

***Tip:*** *If you are already deep in the UiPath ecosystem, AI Units simplify billing. If you are building custom agentic architectures, direct API access gives you more control and often better pricing.*

# 11. Cost Optimization Strategies

## Strategy 1: Model Cascading

Route simple queries to cheap models, complex queries to expensive models.

```
# Model cascading logic
def route_to_model(query_complexity: str):
    if complexity == "simple":
        return "gpt-4o-mini"   # $0.75/1M total
    elif complexity == "medium":
        return "gpt-4.1-mini"  # $2.00/1M total
    else:
        return "gpt-4o"        # $12.50/1M total

# If 70% simple, 20% medium, 10% complex:
# Blended cost is ~60-70% cheaper than using GPT-4o for all
```

### Strategy 2: Semantic Caching

Cache responses for semantically similar queries. Can reduce LLM calls by 30-50% for repetitive workloads.

### Strategy 3: Prompt Optimization

Shorter prompts = fewer input tokens. A prompt reduced from 2,000 to 1,000 tokens saves 50% on input costs across all calls.

### Strategy 4: Batch API

For non-real-time workloads (evaluation, data processing), use OpenAI's Batch API for a 50% discount.

### Strategy 5: Hybrid PAYG + PTU

Reserve PTUs for your baseline load. Use PAYG for burst/overflow traffic. This optimizes cost while maintaining SLAs for core workload.

```
# Hybrid approach
Baseline: 100K TPM sustained -> Reserve 40 PTUs
Burst (2-3 hours/day): Additional 50K TPM -> PAYG overflow
Night batch processing -> Batch API (50% off)

Total cost = PTU (fixed) + PAYG (variable) + Batch (discounted)
```

# 12. Decision Framework

Use this decision tree to select the right pricing model for your use case.

### Decision Questions

- **Q1:** Is your monthly LLM spend > $10,000/month on a single model? If No: use PAYG.
- **Q2:** Is your traffic pattern steady (not spiky)? If No: use PAYG.
- **Q3:** Do you have a latency SLA < 5 seconds P95? If Yes: consider PTU.
- **Q4:** Can you commit for 1 year for 35% savings? If Yes: Annual PTU.
- **Q5:** Is your usage > 60% of PTU capacity during work hours? If Yes: PTU is cost-effective.

| Monthly LLM Spend | Traffic Pattern | Latency Needs | Recommendation |
|---|---|---|---|
| < $1,000 | Any | Any | PAYG |
| $1K - $10K | Variable | Flexible | PAYG |
| $1K - $10K | Steady | Strict SLA | PAYG + evaluate PTU |
| $10K - $100K | Variable | Flexible | PAYG + caching + cascading |
| $10K - $100K | Steady | Strict SLA | PTU (annual reservation) |
| > $100K | Any | Any | Hybrid PTU + PAYG + Batch |

**Key Takeaway:** Most enterprises should start with PAYG, implement cost optimizations (caching, cascading, prompt optimization), and only move to PTU when monthly spend on a single model consistently exceeds $10K and latency predictability is a hard requirement.